

- 4 (a) (i) Gravitational field strength at a point is defined as the gravitational force per unit mass acting on a small test mass placed at that point. **B1**

- (ii) **M1**

$$\frac{GMm}{R^2} = mg$$

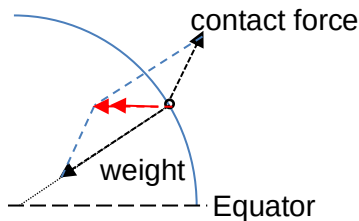
$$M = \frac{gR^2}{G} = \frac{(9.81)(6400 \times 10^3)^2}{6.67 \times 10^{-11}}$$

M1

$$= 6.02 \times 10^{24} \text{ kg}$$

A0

- (b) (i) 1.



The double-headed arrow shows the direction of the centripetal force (ie the resultant force) on the boy as he rotates about the Earth's axis. **A1**

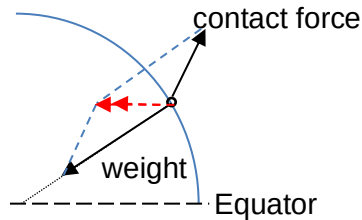
$$2. F = ma = m\omega^2 r = m \frac{4\pi^2}{T^2} r = 50 \times \frac{4\pi^2}{(24 \times 3600)^2} (6\,400\,000 \times \cos 35.6^\circ)$$

M1

$$= 1.38 \text{ N}$$

A1

(ii)



- Weight pointing towards the centre of the Earth **B1**
- Contact force by ground on person comprises of normal contact force and friction. Accept if normal contact force and friction force are drawn separately. *cannot be along the same line as weight and its length cannot be much longer than that for the weight **B1**

We are drawing a free body diagram and so the inclusion of the resultant force (i.e. centripetal force) parallel to the plane of equator and pointing towards the axis of rotation is accepted – deduct 1 mark

Wrong or insufficient labelling – deduct 1 mark

- (ii) At the North pole, the rotational effect of the Earth would be negligible, (thus no centripetal force here) **B1**

the normal contact force that the boy exerted on the scale at the North pole is thus numerically the same as his true weight. **B1**

At the current location (35° north of equator), the reading on the weighing scale is smaller than the true weight due to the rotational effect of the Earth on the boy. Part of the gravitational force is given to the centripetal force which allows for the boy to rotate about the Earth's axis.

Note the forces on the boy

- At the North pole,

Net force = $W - N = 0$, thus $N = W$ (Normal force numerically equals to true weight as there is no centripetal force)

- At the equator,

Net force = $W - N' = \text{Centripetal Force}$, thus $N' = W - F_c$

Or $W = N' + F_c$

The normal force is less at the equator as some of the gravitational force on the object is used to provide for the centripetal force.